



Ultra AI Carousel Engine (Cloudflare + FB Carousel Automation)

Features

- Procedural premium backgrounds
 - SVG → PNG rendering
 - Instagram/Facebook carousel slides
 - Auto typography fit engine
 - Highlighted keywords
 - Glassmorphism
 - Decorative SVG elements
 - Emoji accents
 - Auto CTA
 - Multi-slide AI storytelling
 - Facebook carousel upload automation
 - KV dedupe system
 - Cloudflare Cron automation
 - Fully code-generated visuals
 - No stock-image dependency
-



Folder Structure

```
project/
├── src/
│   ├── worker.js
│   ├── ai.js
│   ├── themes.js
│   ├── slides.js
│   ├── typography.js
│   ├── renderer.js
│   ├── facebook.js
│   ├── backgrounds.js
│   ├── utils.js
│   └── constants.js
├── package.json
├── wrangler.toml
└── fonts/
    └── NotoSansDevanagari.ttf
```

wrangler.toml

```
name = "ultra-carousel-engine"
main = "src/worker.js"
compatibility_date = "2025-01-01"

[[kv_namespaces]]
binding = "CACHE"
id = "YOUR_KV_ID"

[vars]
PAGE_ID = "YOUR_PAGE_ID"

[triggers]
crons = ["0 */3 * * *"]
```

package.json

```
{
  "name": "ultra-carousel-engine",
  "type": "module",
  "dependencies": {
    "@resvg/resvg-js": "^2.6.2"
  }
}
```

src/constants.js

```
export const WIDTH = 1080;
export const HEIGHT = 1350;
```

src/themes.js

```
export const THEMES = {
  psychology: {
    bg1: '#0f172a',
    bg2: '#1e293b',
    accent: '#38bdf8',
    accent2: '#8b5cf6',
    text: '#ffffff',
    sub: '#cbd5e1'
  },

  finance: {
    bg1: '#111111',
    bg2: '#1c1917',
    accent: '#fbbf24',
    accent2: '#f59e0b',
    text: '#ffffff',
    sub: '#e7e5e4'
  },

  motivation: {
    bg1: '#3b0764',
    bg2: '#581c87',
    accent: '#ec4899',
    accent2: '#f97316',
    text: '#ffffff',
    sub: '#f5d0fe'
  }
};
```

src/ai.js

```
export async function generateCarouselStory(env, topic) {

  const prompt = `
Generate a 5-slide viral Hindi carousel.

Topic: ${topic}

Rules:
- Hindi only
```

- Emotional + curiosity based
- Each slide max 15 words
- Strong hook
- Practical insight
- CTA ending

Return JSON array:

```
[
  {
    "type": "hook",
    "text": "...",
    "highlight": ["..."],
    "emoji": "🔥"
  }
];

const res = await fetch(
  'https://api.groq.com/openai/v1/chat/completions',
  {
    method: 'POST',
    headers: {
      Authorization: `Bearer ${env.GROQ_API_KEY}`,
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({
      model: 'llama-3.3-70b-versatile',
      temperature: 0.8,
      max_tokens: 1000,
      messages: [{ role: 'user', content: prompt }]
    })
  }
);

const json = await res.json();

const text = json.choices[0].message.content
  .replace(/``json/g, '')
  .replace(/``/g, '')
  .trim();

return JSON.parse(text);
}
```

src/typography.js

```
export function getFontSize(text) {

  const len = text.length;

  if (len < 30) return 82;
  if (len < 50) return 72;
  if (len < 70) return 62;
  if (len < 90) return 54;

  return 46;
}

export function wrapText(text, maxChars = 15) {

  const words = text.split(' ');
  const lines = [];

  let line = '';

  for (const word of words) {

    if ((line + word).length > maxChars) {
      lines.push(line.trim());
      line = word + ' ';
    }
    else {
      line += word + ' ';
    }
  }

  if (line.trim()) lines.push(line.trim());

  return lines;
}
```

src/backgrounds.js

```
export function meshBackground(theme) {

  return `
```

```

<defs>

  <radialGradient id="g1">
    <stop offset="0%" stop-color="${theme.accent}" stop-opacity="0.45"/>
    <stop offset="100%" stop-color="${theme.bg1}" stop-opacity="0"/>
  </radialGradient>

  <radialGradient id="g2">
    <stop offset="0%" stop-color="${theme.accent2}" stop-opacity="0.45"/>
    <stop offset="100%" stop-color="${theme.bg2}" stop-opacity="0"/>
  </radialGradient>

  <filter id="blur">
    <feGaussianBlur stdDeviation="60"/>
  </filter>

</defs>

<rect width="100%" height="100%" fill="${theme.bg1}"/>

<circle cx="250" cy="250" r="350"
  fill="url(#g1)"
  filter="url(#blur)"/>

<circle cx="850" cy="1000" r="350"
  fill="url(#g2)"
  filter="url(#blur)"/>
`;
}

```



src/slides.js

```

import { WIDTH, HEIGHT } from './constants.js';
import { getFontSize, wrapText } from './typography.js';
import { meshBackground } from './backgrounds.js';

export function buildSlide(slide, theme) {

  const fontSize = getFontSize(slide.text);

  const lines = wrapText(slide.text);

  let y = 0;

```

```

const textSvg = lines.map(line => {

  const output = `
    <text
      x="100"
      y="${y}"
      font-size="${fontSize}"
      fill="white"
      font-family="Arial"
      font-weight="900"
    >
      ${line}
    </text>
  `;

  y += fontSize * 1.5;

  return output;

}).join('');

return `

<svg
  xmlns="http://www.w3.org/2000/svg"
  width="${WIDTH}"
  height="${HEIGHT}"
>

  ${meshBackground(theme)}

  <rect
    x="60"
    y="100"
    width="960"
    height="1150"
    rx="48"
    fill="rgba(255,255,255,0.06)"
    stroke="rgba(255,255,255,0.1)"
  />

  <text
    x="100"
    y="200"
    font-size="72"
  >
    ${slide.emoji}

```

```
    </text>

    <g transform="translate(0,400)">
      ${textSvg}
    </g>

    <text
      x="100"
      y="1180"
      font-size="30"
      fill="white"
      opacity="0.8"
    >
      @akpage
    </text>

  </svg>
`;
}
```

src/renderer.js

```
import { Resvg } from '@resvg/resvg-js';

export async function svgToPng(svg) {

  const resvg = new Resvg(svg, {
    fitTo: {
      mode: 'width',
      value: 1080
    }
  });

  const png = resvg.render();

  return png.asPng();
}
```




src/facebook.js

```
export async function uploadPhoto(pageId, token, imageBuffer) {

  const form = new FormData();

  form.append('published', 'false');
  form.append('access_token', token);

  form.append(
    'source',
    new Blob([imageBuffer], { type: 'image/png' }),
    'slide.png'
  );

  const res = await fetch(
    `https://graph.facebook.com/v22.0/${pageId}/photos`,
    {
      method: 'POST',
      body: form
    }
  );

  return await res.json();
}

export async function createCarouselPost(pageId, token, mediaIds, caption) {

  const attached_media = mediaIds.map(id => ({
    media_fbid: id
  }));

  const form = new URLSearchParams();

  form.append('message', caption);
  form.append('access_token', token);

  attached_media.forEach((m, i) => {
    form.append(`attached_media[${i}]`, JSON.stringify(m));
  });

  const res = await fetch(
    `https://graph.facebook.com/v22.0/${pageId}/feed`,
    {
      method: 'POST',
      body: form
    }
  );
}
```

```

    }
  );

  return await res.json();
}

```

src/utils.js

```

export function randomItem(arr) {
  return arr[Math.floor(Math.random() * arr.length)];
}

export async function dedupe(env, key) {

  const exists = await env.CACHE.get(key);

  if (exists) return true;

  await env.CACHE.put(
    key,
    '1',
    { expirationTtl: 86400 * 30 }
  );

  return false;
}

```

src/worker.js

```

import { THEMES } from './themes.js';
import { randomItem, dedupe } from './utils.js';
import { generateCarouselStory } from './ai.js';
import { buildSlide } from './slides.js';
import { svgToPng } from './renderer.js';
import {
  uploadPhoto,
  createCarouselPost
} from './facebook.js';

const TOPICS = [

```

```

    'क्यों 90% लोग सफल नहीं हो पाते?',
    'दिमाग की वो आदत जो आपकी जिंदगी बदल सकती है',
    'क्या आपका दिमाग आपको पीछे रोक रहा है?',
    'अमीर लोग बच्चों को कौन सा सीक्रेट सिखाते हैं?'
  ];

export default {

  async fetch(req, env) {

    const result = await run(env);

    return Response.json(result);
  },

  async scheduled(event, env, ctx) {
    ctx.waitUntil(run(env));
  }
};

async function run(env) {

  const topic = randomItem(TOPICS);

  const duplicate = await dedupe(env, topic);

  if (duplicate) {
    return {
      skipped: true,
      reason: 'duplicate topic'
    };
  }

  const slides = await generateCarouselStory(env, topic);

  const theme = THEMES.psychology;

  const mediaIds = [];

  for (const slide of slides) {

    const svg = buildSlide(slide, theme);

    const png = await svgToPng(svg);

    const upload = await uploadPhoto(
      env.PAGE_ID,

```

```
    env.PAGE_ACCESS_TOKEN,  
    png  
  );  
  
  mediaIds.push(upload.id);  
}  
  
const post = await createCarouselPost(  
  env.PAGE_ID,  
  env.PAGE_ACCESS_TOKEN,  
  mediaIds,  
  'क्या आप इससे सहमत हैं? 📌'  
);  
  
return {  
  success: true,  
  post  
};  
}
```

Add Secrets

```
wrangler secret put PAGE_ACCESS_TOKEN  
wrangler secret put GROQ_API_KEY
```

Deploy

```
npm install  
wrangler deploy
```

Next-Level Upgrades

Add:

- AI viral scoring
- animated SVG export
- video reels rendering

- subtitle motion engine
 - multilingual generation
 - AI hook optimizer
 - CTA optimizer
 - A/B testing
 - trend scraper
 - auto hashtags
 - Telegram crossposting
 - Instagram support
 - Pinterest automation
-



Best Niches

- psychology
 - finance
 - motivation
 - spirituality
 - productivity
 - dark facts
 - business mindset
 - relationship psychology
-



Final Result

Tumhara system automatically:

- viral topic choose karega
- 5-slide story generate karega
- premium visuals banayega
- typography fit karega
- FB carousel post karega
- duplicates avoid karega
- aesthetic branded output dega

Completely automated AI media engine.